

Mechanistically Examining a Financial Agent

We tested whether the model’s internal picture of the market is real, where it comes from, and whether it actually drives the trading choice. Six findings, one honest verdict.

DATE	MODEL	SCOPE	STATUS
April 2026	Qwen3-30B-A3B (MoE)	6 findings · 9 phases	Validated through causal restoration test

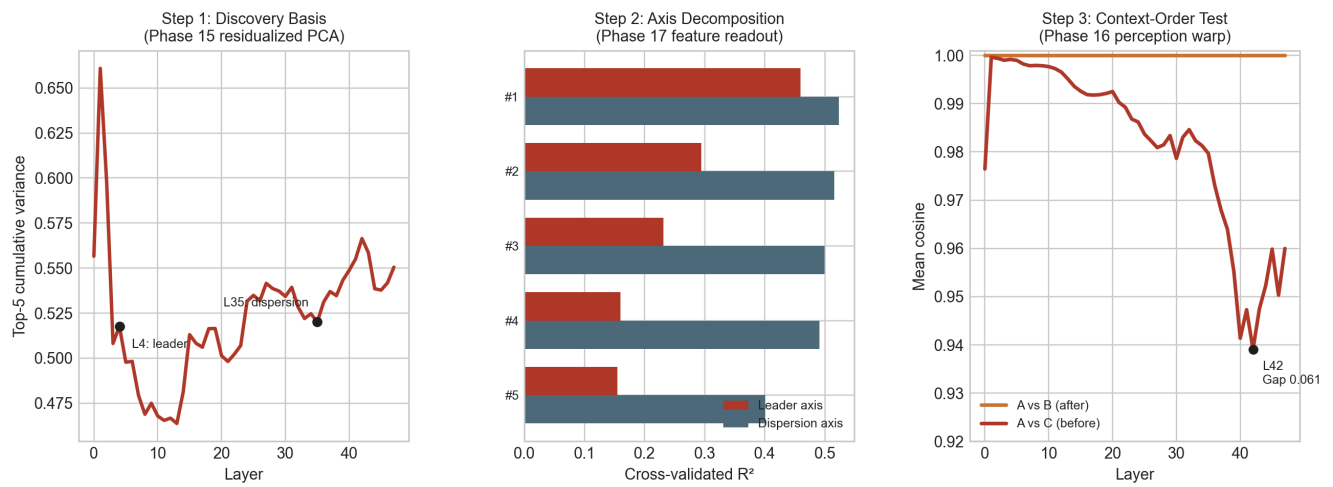
BOTTOM LINE

The model builds a precise, structured internal picture of the market, and prior context measurably shapes that picture. The two specific internal signals we isolated are real and selective — but on their own, they only partially explain the model’s final trading choice.

MARKET FIDELITY	DATA FIDELITY	RELATIONAL ROBUSTNESS	SELECTIVITY	RESTORATION
> 0.99 R^2		$\approx 20\times$	12 / 12	1. 4.2 pp
Every tested factor recoverable from internal state		Pairwise vs single-asset stability under reformatting	Targeted edits beat matched random edits	Choice agreement shift on the strongest signal

WHY THIS MATTERS

We extracted the model’s hidden activations as it processed trading-style prompts and ran six progressively tighter tests on what we found there — from representation discovery and format decontamination through context-sensitivity, intervention, robustness, and causal restoration. Each step asks a sharper version of the same question: what does the model represent internally, and does any of it actually drive the trade?

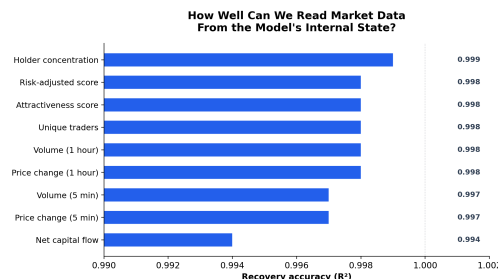


The full study at a glance: discovery basis (Phase 15) → axis decomposition (Phase 17) → context-order test (Phase 16).

Finding 1 — The Model Preserves Market Data With High Fidelity

Linear probes on early-layer activations recover every market metric we tested with near-perfect accuracy. Price changes at two timescales, trading volume at two timescales, net capital flow, unique trader counts, holder concentration, and the composite scores all decode at $R^2 > 0.994$. The model does not compress, summarize, or hallucinate the input — downstream decisions are grounded in the actual numbers it was given.

High points. Holder concentration **0.999** 1-hour price, 1-hour volume, unique traders, and both composite scores **0.998** net capital flow **0.994**. Whatever the model is doing later, it is doing it on top of a faithful copy of what it read.

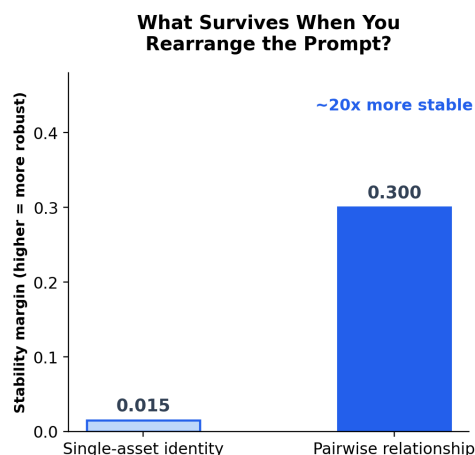


Recovery R^2 for nine market factors. All > 0.994 .

Finding 2 — The Representation Is Relational, Not Row-By-Row

When the model evaluates a token, it does so relative to the full field. We reformatted prompts (rearranging row order, swapping ticker symbols, altering text style) and watched what survived. Single-asset identity broke easily under layout changes. Pairwise relationships between assets were approximately **20×** more robust on the same prompt set.

Scale. 384 controlled prompt variations across four scenario families, multiple formatting styles, and three magnitude scales. The model's internal market is encoded as a structure of comparisons, not a list of independent rows.

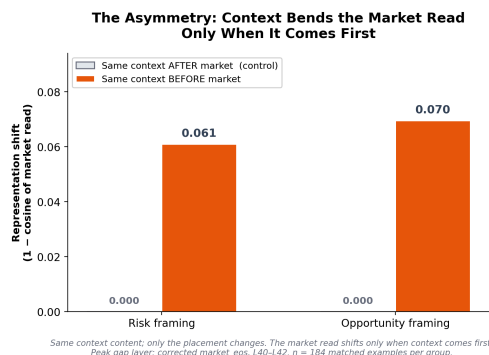


Single-asset vs pairwise stability under reformatting.

Finding 3 — Pre-Market Context Shapes the Market Read

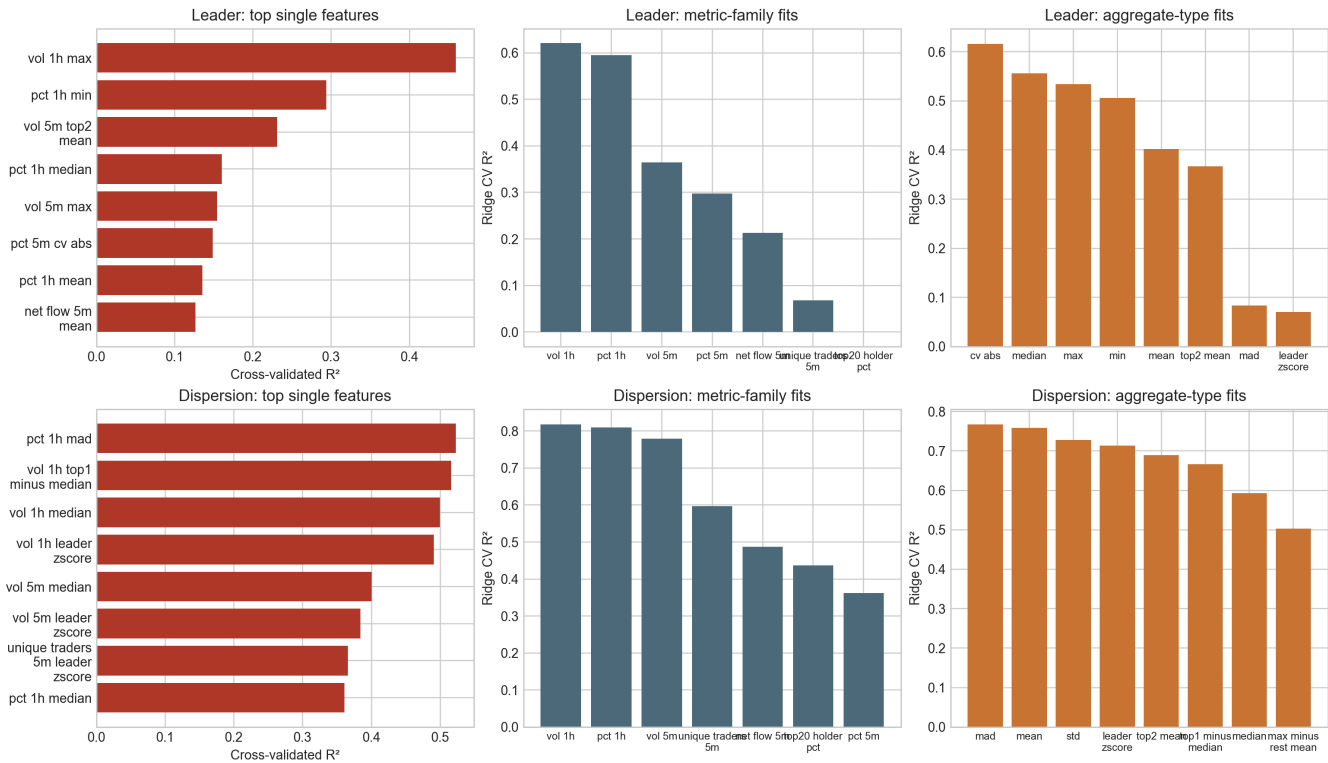
Identical markets land differently when the framing comes first. We held the market section constant and moved the same context block either before or after it — **same words, only the placement changes**. When context appeared after the market, the internal market read was unchanged (**0.000** shift). When it appeared before the market, the read shifted: risk framing **0.061** (peak at L42), opportunity framing **0.070** (peak at L40).

The asymmetry is the result. Not “more text moves the representation” — only context that arrives in time to act as a lens does. Confirmed on the full five-level risk ladder: the base coordinate system stays nearly perfect ($R^2 > 0.995$ cross-context) while the **interpretation** shifts in a structured, ladder-appropriate way.



Phase 16: same context, only the placement changes.

Finding 4 — Two Named Internal Signals



Top single features, metric-family fits, and aggregate-type fits for the two axes. Leader (red) is dominated by **top-asset** features; Dispersion (teal) by **spread** features.

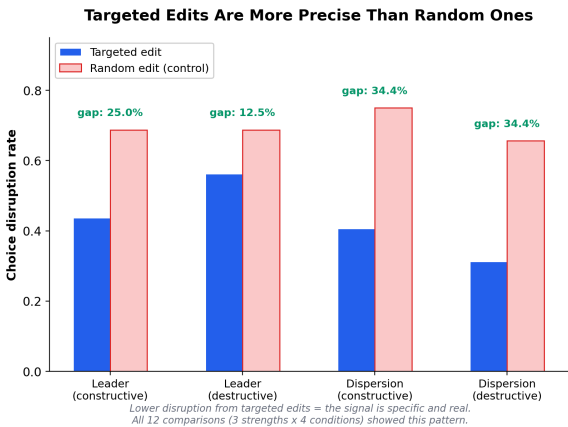
After controlling for prompt-formatting artifacts (sequence length, character count, asset count), statistical decomposition of the model’s market-section activations identifies two recurring internal patterns. **Leader** (early layers) tracks the standout asset: best single feature R^2 **0.46**, best pair R^2 **0.67**. **Dispersion** (later layers) tracks how uneven the market is: best single feature R^2 **0.52**, best pair R^2 **0.84**. Both collapsed below R^2 **0.06** on shuffled-data controls — confirming they reflect market content, not noise — and both decode from features a human trader could also read off the screen.

Finding 5 — The Signals Are Real (Selectivity Test)

We edited the model’s internal state at each signal site and measured behavioral disruption against matched random edits of the same magnitude. In every tested condition, the targeted edit produced **less** collateral disruption than the matched random one — the signature of a meaningful internal feature, not a generic perturbation effect.

Condition	Targeted	Random	Gap
Leader, constructive	43.8%	68.8%	25.0 pp
Leader, destructive	56.3%	68.8%	12.5 pp
Dispersion, constructive	40.6%	75.0%	34.4 pp
Dispersion, destructive	31.3%	65.6%	34.4 pp

Across three strengths $\times$ four conditions, **all 12 comparisons** showed the same pattern. Both signals carry specific, non-redundant information.



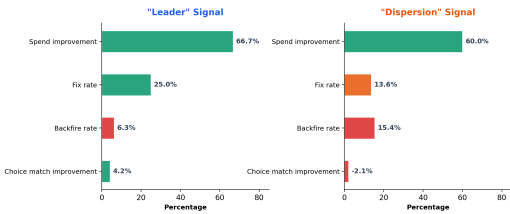
Targeted vs matched random edits at strength 1.0.

Finding 6 — Restoration: The Honest Causal Test

Earlier tests damaged signals and watched for harm. Restoration reverses the logic: take each signal from a donor scenario, transplant it into a base scenario, and measure whether the model shifts toward the donor’s choice. **Gold standard for causal claims, and where most appealing stories fall apart.**

Signal	Δ choice agreement	Fix rate	Backfire
Leader	+4.2 pp	25.0%	6.3%
Dispersion	-2.1 pp	13.6%	15.4%

Leader is a real but partial causal handle: modest positive shift, fix rate 4× the backfire rate, spend improvement on 66.7% of cases. **Dispersion** failed: backfire exceeded fix rate. 48 paired scenarios per signal.



Restoration outcomes for the two signals.

What Held Up

Claim	Support	Why this rating
The model preserves raw market data internally with high fidelity.	Strong	All 9 tested factors decode at $R^2 > 0.994$. Consistent across prompt variants.
The model’s market understanding is comparative, not row-by-row.	Strong	Pairwise relationships 20× more stable than single-asset identity. 384 prompt variations.
Pre-market context shapes how the model reads the market.	Strong	Asymmetry: 0.000 shift when context comes after, 0.061–0.070 when before. Confirmed across full risk ladder.
The two named signals (Leader, Dispersion) reflect genuine market content.	Strong	Targeted edits beat matched random edits in 12/12 comparisons. Both collapse below R^2 0.06 on shuffled controls.
The Leader signal partially drives the model’s trading choice.	Partial	Restoration produces +4.2 pp shift toward donor scenario, fix rate 4× the backfire rate. Real but modest.
The Dispersion signal does not meaningfully drive the trading choice.	Negative	Restoration backfire (15.4%) exceeds fix rate (13.6%). The signal is real (Finding 5), but not a causal handle.

WHAT WE DON’T KNOW

- The single decisive cause of any given trade is still not isolated. Our strongest causal handle (Leader) explains a slice, not the whole picture.
- The integration point where market perception meets user-specific factors — portfolio, constraints, strategy — is where the rest of the explanation lives.
- Real but readable internal structure does not automatically translate to a behavioral lever. Restoration is the only test that distinguishes the two.

VERDICT

The model builds a meaningful, structured internal summary of the market, and that summary is influenced by prior context. We can identify recurring internal signals that line up with visible market features. One of those signals (Leader) partially steers the trading choice. The other (Dispersion) does not. Narrower is not weaker — these are the claims the evidence actually supports.

TAKEAWAY

The model builds a real, structured market picture, and one of the two named signals partially drives the trading decision. The other doesn't.

WHY IT MATTERS

First end-to-end mechanistic interpretability study on a deployed financial agent. Causal interp scales to production systems, not just toy models.

WHAT'S NEXT

Apply the same playbook to detect **conflict in trading policies** — pinning down where competing policy signals live and which ones actually drive action.

Appendix

A.1 • Model and Scale

Model	Qwen3-30B-A3B (MoE)
Total parameters	30 B
Active per token	3 B
Layers examined	48
Expert routing	top-8 of 60 per MoE
Synthetic scenarios / experiment	184–920
Real inference logs (reference)	203,292
Real activation captures	11,579
Intervention conditions	12 ($2 \times 2 \times 3$)
Paired scenarios / condition	48
Capture infra	Modal · A100-80GB
Inference engine	vLLM + capture hooks

A.2 • Methodology

- **Linear probing.** Ridge regression, L1-regularized logistic regression, and SGD with balanced class weights, on grouped held-out splits (no row leakage across train/test).
- **Activation capture.** Full-sequence pooled into row, market-mean, market-eos, and downstream-section states. Format-confound residualization removes the slice predictable from sequence length, character count, and asset count.
- **Decomposition.** PCA on residualized market-section activations; the two dominant directions are regressed against human-readable market features to assign Leader and Dispersion labels.
- **Intervention.** Source-driven swap_components patching at the residual stream — source row's market-span activations are averaged and the selected coefficients are inserted into the base prompt's market span.
- **Restoration.** Matched-pair denoise design: the lower-valued member of each pair is the base, the higher-valued is the source. The patch tries to move the base toward the source's behavior.

A.3 • Validation and Controls

- **Format controls.** Layout permutations, ticker swaps, row reordering, and style changes confirm findings reflect market content rather than surface text patterns.
- **Matched random controls.** Every targeted intervention is paired with a random edit of the same magnitude at the same site. The 12/12 selectivity result is defined against this control.
- **Shuffle baselines.** Both named signals collapse below R^2 0.06 on shuffled inputs, ruling out spurious statistical relationships.
- **Bootstrap CIs.** All behavioral metrics include bootstrap confidence intervals at 2,000 resamples.
- **Decode-validity check.** For Phase 21 restoration, all 48/48 base and source rows on both axes produced parsed tool calls with finish_reason=stop and zero max_tokens cap hits — the action surface is observable, not truncated.

A.4 • Path-Validation Status

We ran a Phase 22 path-validation test on the Leader axis: an early lesion (project_out at L4) followed by a downstream rescue (swap_components at L35) using source-side coefficients. **The patch path itself worked** — every row produced patch diagnostics, no rows were skipped, both lesion and rescue applied cleanly. **But** the lesioned model did not reliably produce parsed tool calls at scale, so the action-choice readout was too sparse to interpret causally. This is not a refutation of any finding above; it is where the next round of work begins. Two natural follow-ups: (a) a denser behavioral readout regime that survives the lesion, and (b) moving from span-mean coefficient swapping toward exact source-activation transplantation.